

USER MANUAL

BUC Redundant Switch System (950-2000 MHz) 1:1
RSBUCWC



TABLE OF CONTENTS

Revision History	3
1. Set Up	3
1.1 Setup Overview.....	3
1.2 Minimum Hardware and Software Requirements	3
1.3 Software Installation.....	3
1.4 Connection of the PC and the RCU.....	5
2. Redundant Control System.....	8
2.1 System Configuration.....	8
2.2 Redundant System Operate Flow Chart	9
3. Monitoring and Control Program.....	9
3.1 RCS_V3.xx Program Operation	9
3.2 Description of RCS_V3.xx Program.....	10
3.3 Operation	12
4. Web-Based Monitoring and Control.....	21
4.1 Web-Based RCS Operation	21
4.2 Description of Web-based Monitoring and Control	22
4.3 Operation	24
5. Customer Service.....	38
5.1 Warranty Repair	38
5.2 Exceptions to Warranty	38
5.3 Other Repairs	39
5.4 Contact Information.....	39

Revision History

REVISION	DESCRIPTION
A	Initial Release

1. Set Up

1.1 Setup Overview

1. Install the software on your PC.
2. Connect the RS-232/RS-485/Ethernet cable to your PC and connect the other end of the cable to the RCU.
3. Power on the RCU.
4. Power on the BUCs.
5. Start the software installation.
6. Complete the Redundant Control System initial startup and connection configuration.

1.2 Minimum Hardware and Software Requirements

- Windows based desktop or laptop PC.
 - Operating System: Windows® XP, Windows® 7, Windows® 8, Windows® 10 and Windows® 11.
- The USB stick included contains the program file (if applicable).
- The 10 pin of adaptor cable circular connector and cable for M&C function.
- The cable that connects the RCU to the PC.

1.3 Software Installation

1. Insert the USB stick into your PC.
2. Open My Computer and find the USB drive.
3. Double-click **RCS_V3_Setup** folder.
4. Double-click **Volume** folder.
5. Double-click **setup.exe** file to start installation.
6. Wait for the installation progress.

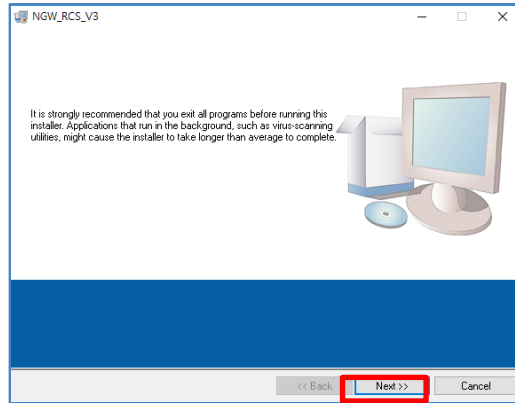


Figure 1.1

- 1) Select the installation directory location then click Next.

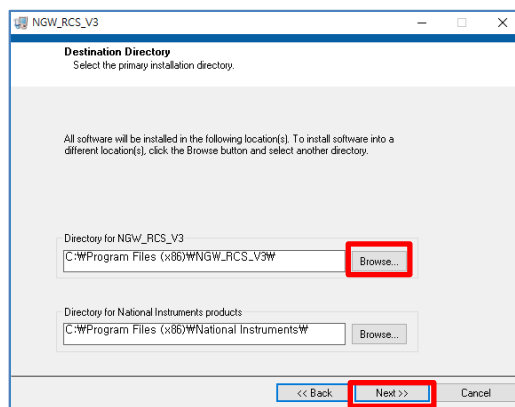


Figure 1.2

- 2) Check I accept and click Next.

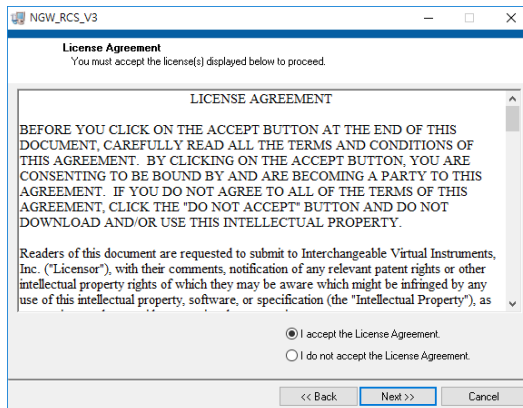


Figure 1.3

- 3) Click Next.

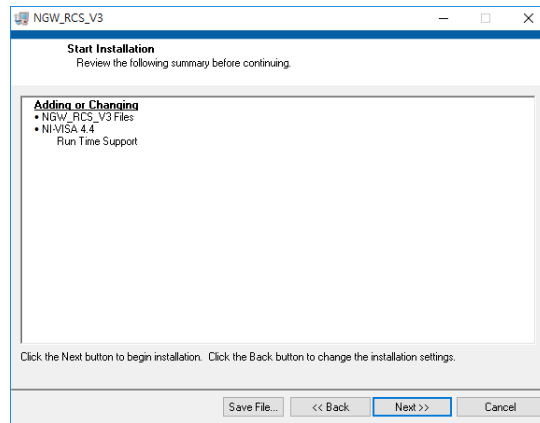


Figure 1.4

- 4) Once the installation is complete, click Finish.

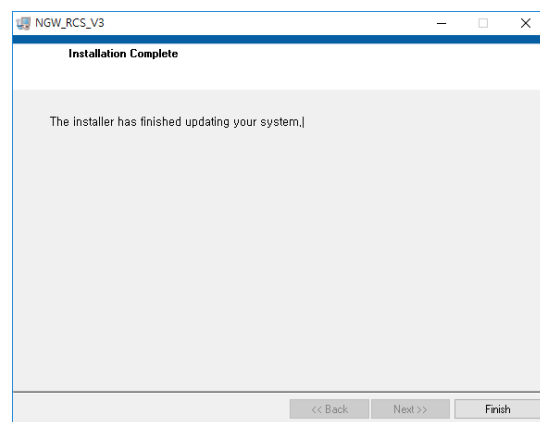


Figure 1.5

- 5) Open the Start menu and click **RCS_V3.xx** to start the program.

1.4 Connection of the PC and the RCU

1. The RCS consists of the BUC A, BUC B, and the RCU.

There are two types of the RCS configuration. The RCS configuration depends on the BUC appearance as below.

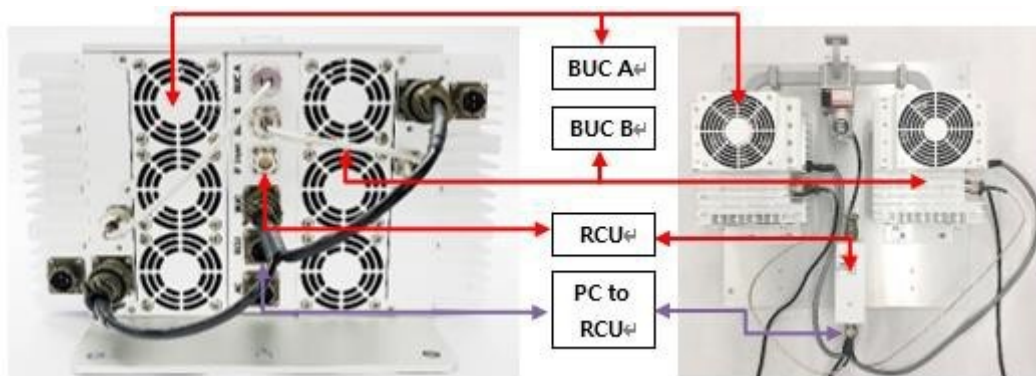


Figure 1.6

2. Connect the RS-232/RS-485 or Ethernet cable to the PC.

If the PC does not have a D-Sub port, a D-Sub to USB adapter will be required.

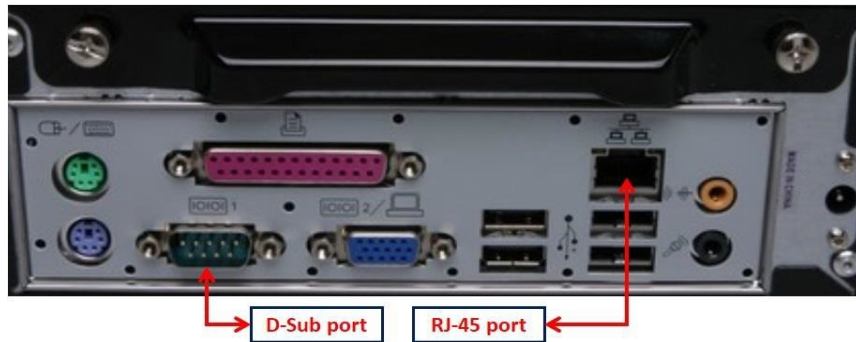


Figure 1.7

3. Connect the 10 pins of adapter cable circular connector to the mating remote M&C port on the RCU and lock the MS connector. Refer to **Figure 1.8** and **Figure 1.9** for connection.

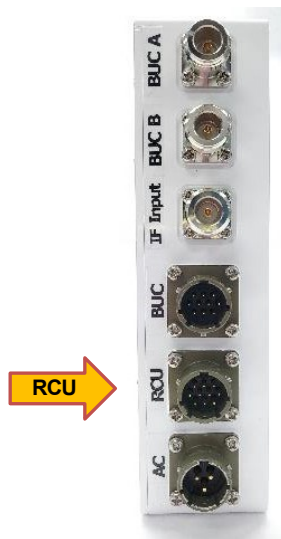
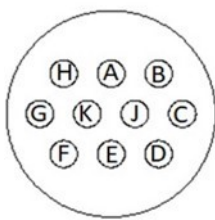


Figure 1.8 - M&C Connector of RCU

- 1) MS connector 10 pins(P) of RCU



MS Connector Pin Number	M&C Interface
A	RS-232 Tx
B	RS-232 Rx
C	Ground
D	NC
E	RS-485 TRx -
F	RS-485 TRx +
G	Ethernet Rx +
H	Ethernet Rx -
K	Ethernet Tx +
J	Ethernet Tx -

Figure 1.9 - Physical Interface of Connector of the RCU

4. Connect the 5 pins or 3 pins of adapter cable circular connector to the mating remote AC port on the BUC and the RCU and lock the MS connector. Refer to **Figure 1.10** and **Figure 1.11** for connection.

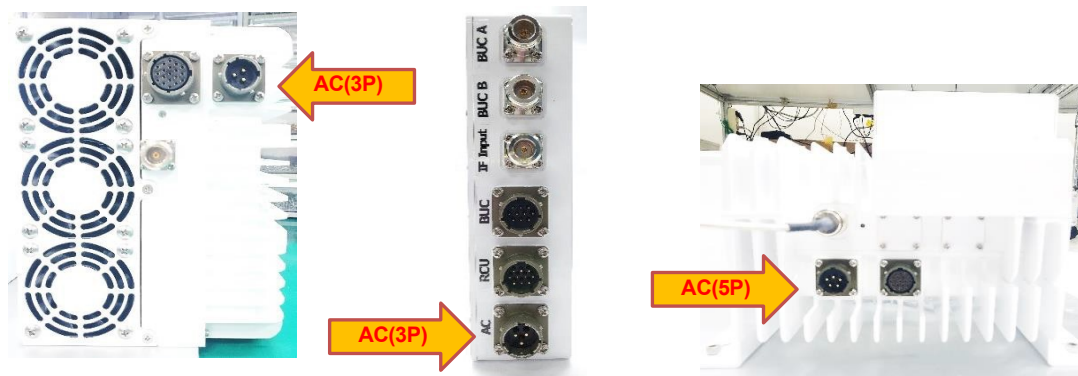
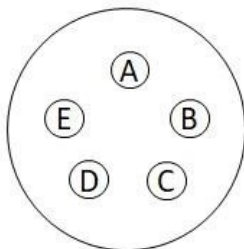


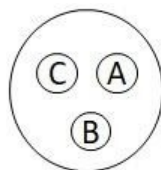
Figure 1.10 - AC Connector of the BUC and the RCU

- 1) MS connector 5 pins (P) of the BUC



MS Connector Pin Number	AC Interface
A	L
B	N
C	G
D	NC
E	NC

- 2) MS connector 3 pin(P) of the BUC and the RCU



MS Connector Pin Number	AC Interface
A	L
B	N
C	G

Figure 1.11 - Physical Interface of Connector of the BUC and the RCU

2.Redundant Control System

2.1 System Configuration

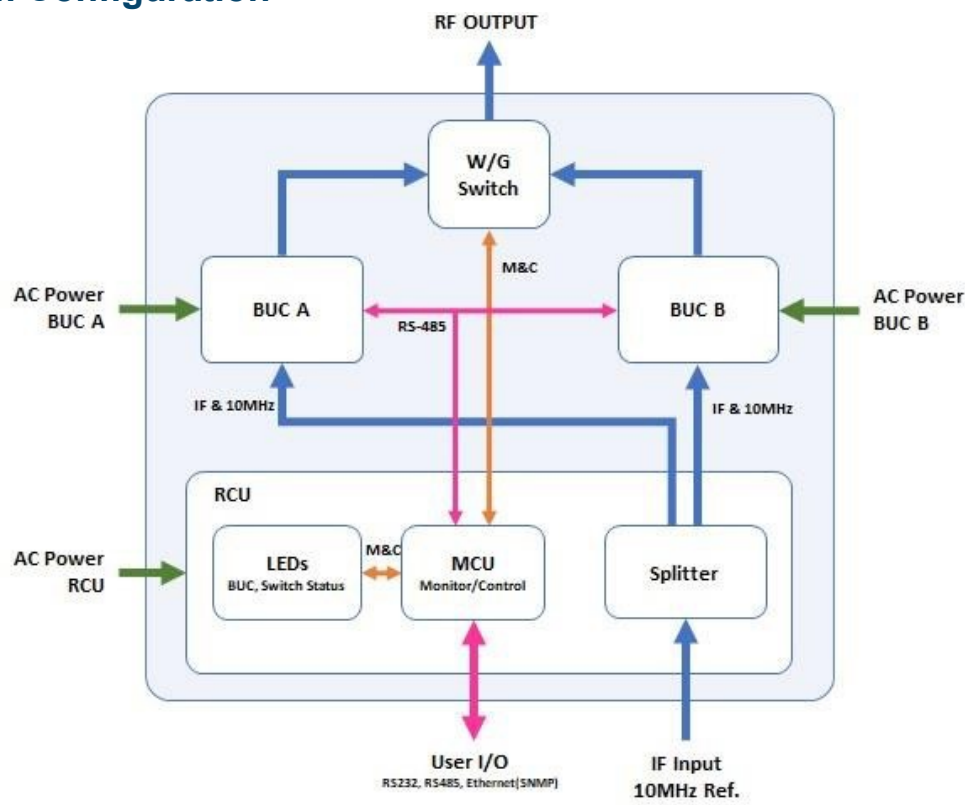


Figure 2.1 - Redundant Control System Configuration

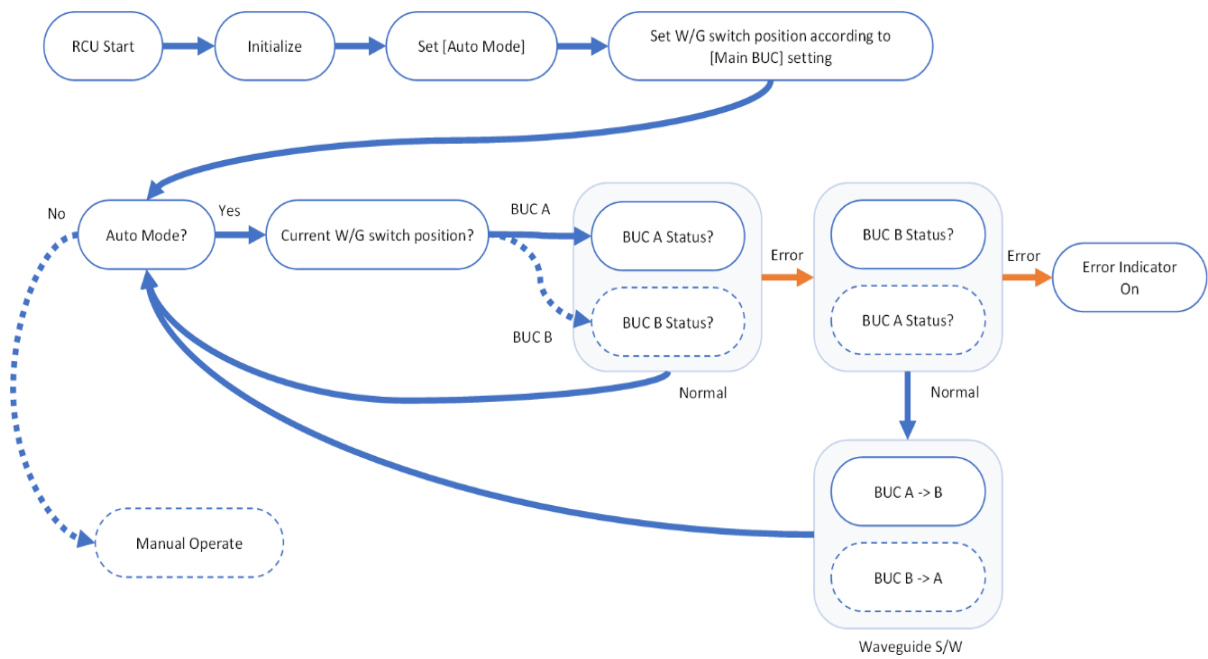


Figure 2.2 - Redundant System Operate Flow Chart

2.2 Redundant System Operate Flow Chart

The following process explains automatic operation of the RCS shown in Figure 6 for the information only so that no specific action is required from the end user.

1. RCU started
2. Initialize
3. Set [Auto Mode]: Default operation is [Auto mode].
4. Set a waveguide (W/G) switch position according to the [Main BUC] setting
5. Check [Auto Mode]
 - 1) In [Manual Mode], the RCS will be operated manually.
 - 2) In [Auto Mode],
 - i. Check the current W/G switch position
 - ii. If the default BUC is [BUC A], check [BUC A] status.
 - iii. If the BUC A status is [Normal], go back to ⑤ Check [Auto Mode].
 - iv. If the BUC A status is [Error], check the BUC B status.
 - v. If the BUC B status is [Normal], the W/G switch position will be changed from the BUC A to the BUC B and go back to ⑤ [Auto Mode] Check.
 - vi. If the BUC B status is [Error], the error indicator of the system will be on and go back ⑤ Check [Auto Mode] to see if the BUC A is [Normal].
 - vii. If the BUC A is [Normal], the W/G switch position will be changed from the BUC B to the BUC A again.

3. Monitoring and Control Program

3.1 RCS_V3.xx Program Operation

1. Plug the power cable to the BUC and double-click **RCS_V3.xx** to start the program.
2. The program will start as shown in **Figure 3.1**.

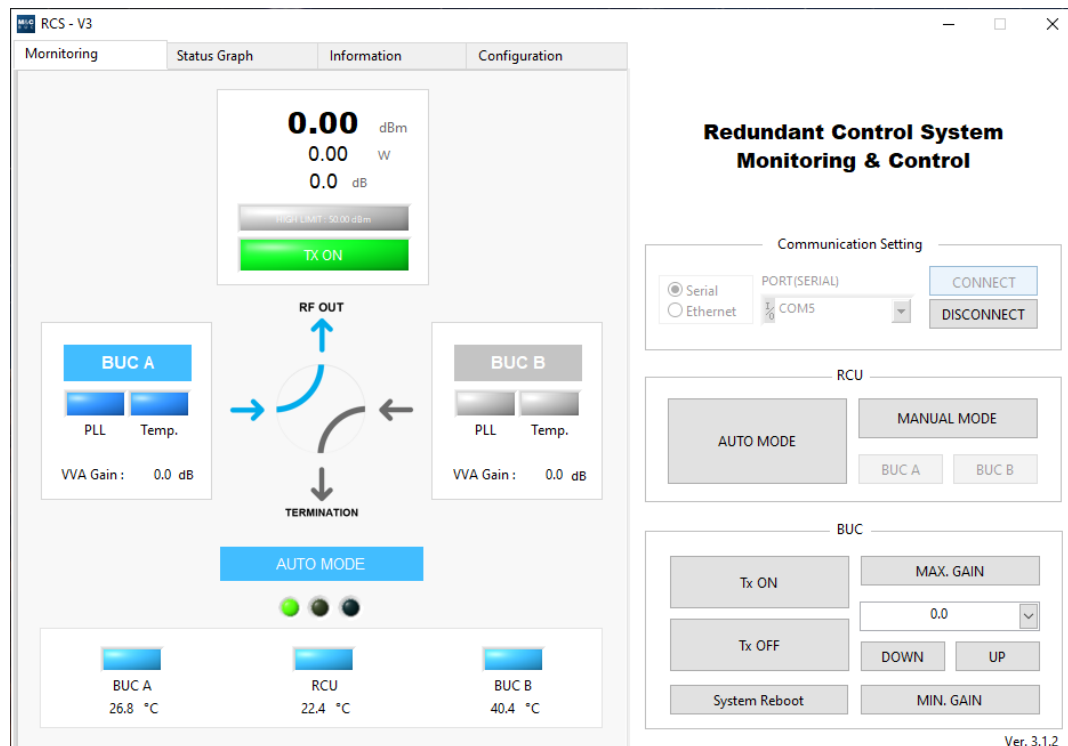


Figure 3.1 – System Diagram

3.2 Description of RCS_V3.xx Program

3.2.1 Communication Setting Window [Serial & Ethernet]

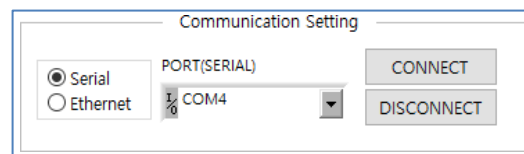


Figure 3.2

- PORT(SERIAL)
From the drop-down menu, the connected port(serial) can be selected.



Figure 3.3

- IP
The default IP address is 192.168.11.110.
- PORT(TCP)
The default PORT (TCP) address is 4200.
- CONNECT / DISCONNECT
These buttons are used to connect or disconnect the communications between the RCU and the RCS_V3.xx program.

3.2.2 RCU Control Window

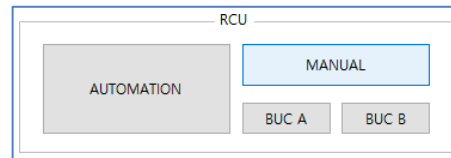


Figure 3.4

1. Automation

If there is a problem with a current operating unit, this mode will switch over to another unit automatically.

2. Manual

If there is a problem with a current operating unit, the operator has to switch over to another unit manually.

3.2.3 BUC Control Window

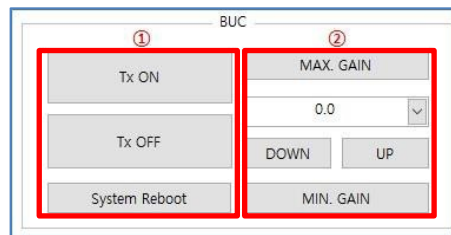


Figure 3.5

1. Control

Tx ON and Tx OFF buttons are used to turn on or off the Tx of the BUC. System reboot button is used to reboot the firmware of the RCU.

2. VVA Gain Setting

These buttons are used to change the gain of the BUC.

3.2.4 RCS Monitoring Window

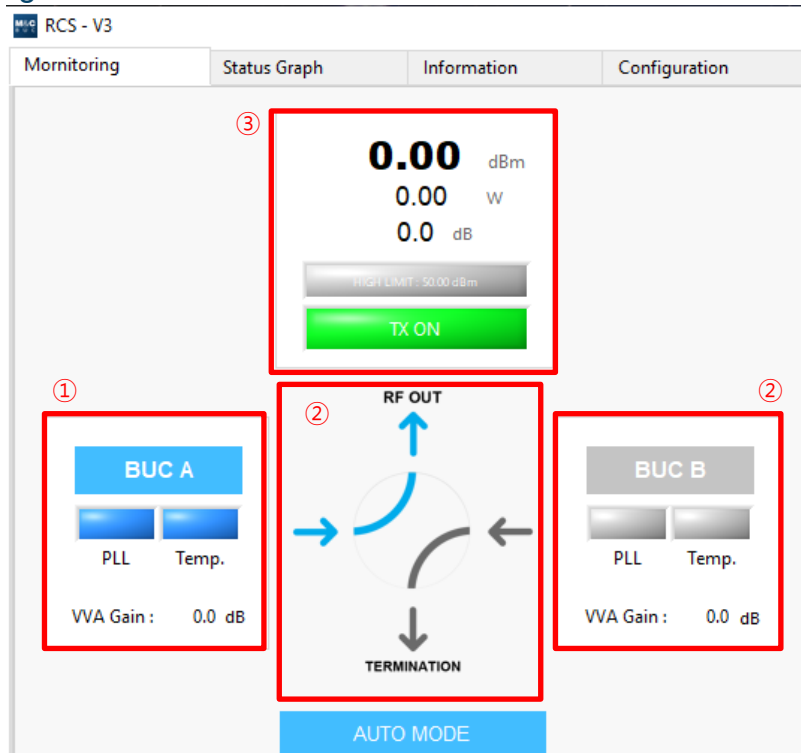


Figure 3.6

1. Main BUC Output Status

This information shows the output power level of the BUC, VVA gain, high limit alert status, and Tx ON/OFF status.

2. Status Indicators

The indicators show Tx ON/OFF status, PLL status, and temperature error status of each BUC.

3. Waveguide Switch

The arrows show the path of the waveguide switch.

3.2.5 LED Indicators & Temperature

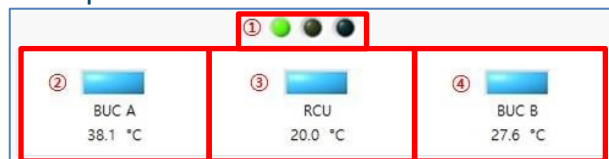


Figure 3.7

1. The indicators show the Tx signal status, Rx signal status, and communication error between Tx and Rx. If they turn red, please check each signal and connection.
2. shows the RCU to the BUC A communication status and temperature.
3. shows the PC to the RCU communication status and temperature.
4. shows the RCU to the BUC B communication status and temperature.

3.3 Operation



Ensure that **RCS_V3.xx** program is installed, and the RCU and the PC are connected.



This program does show the real-time operating status.



If the RCU has any defects, turn off the power of the deactivated BUC before replacing the new RCU.

- 1) Power on the RCU.
- 2) Power on the BUCs.
- 3) Select serial or Ethernet from the Communication Setting window.
 - Select serial
Select the connected port and click CONNECT.
 - Select Ethernet
Setting up IP and PORT (TCP), click CONNECT. The default IP address is 192.168.11.110. The default PORT (TCP) is 4200.
- 4) BUC_A Status Indicator of Tx On/Off will turn into green as shown in **Figure 3.8**.

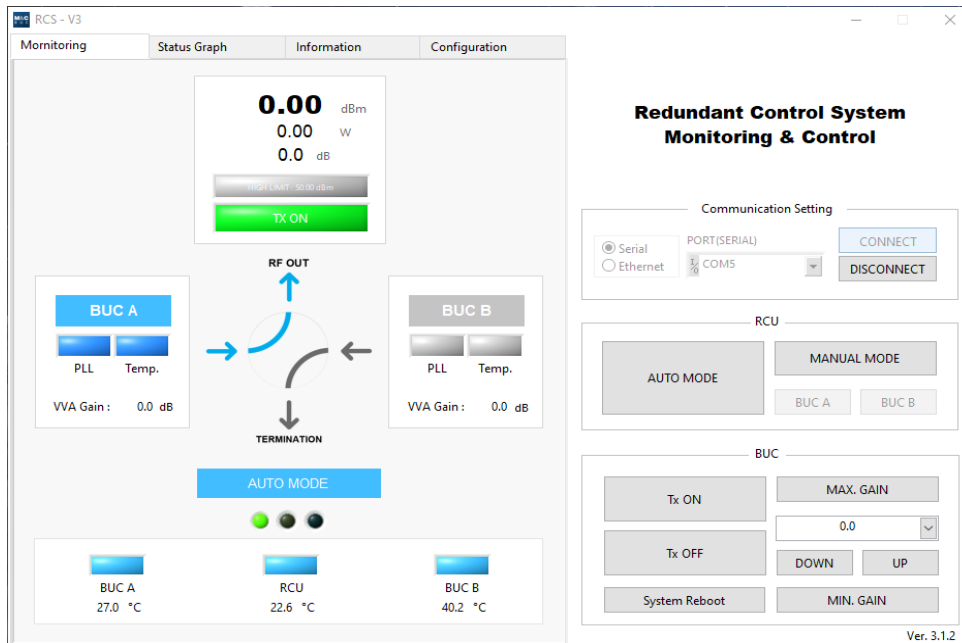


Figure 3.8 - RCS Default Status Without IF Signal

5) Tx ON/OFF Status

Click Tx ON or Tx OFF button in the BUC Window. The indicator of Tx ON/OFF will turn green or off as shown in **Figure 3.9**.

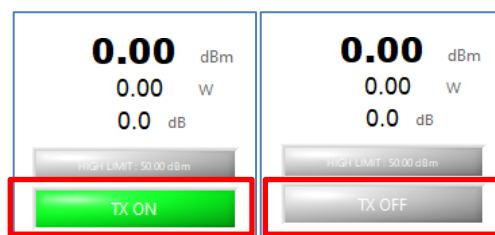


Figure 3.9 - Tx ON and OFF

6) System Reboot

If the RCS monitoring section freezes, click System Reboot button in the BUC Window. The confirmation window will pop up to ask System Reboot the firmware of the RCU as shown in **Figure 3.10**. Click OK button to reboot the firmware of the RCU.

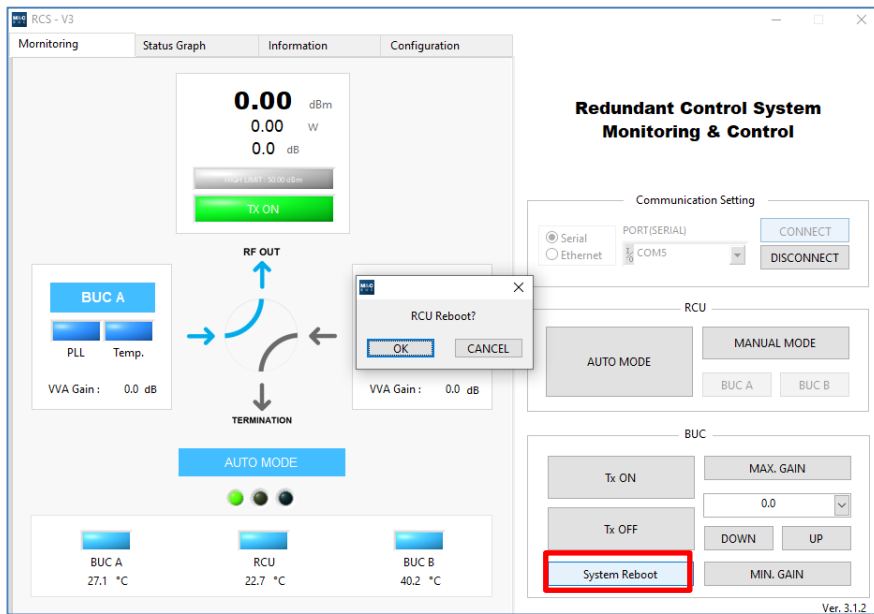


Figure 3.10 - RCU Firmware Reboot

7) VVA Gain Setting

To adjust the gain, the values can be selected from the dropdown menu from -20 dB to 0 dB as shown in **Figure 3.11**. Or it can be adjusted by clicking UP, DOWN, MAX. GAIN or MIN. GAIN buttons. If you click UP or DOWN button, VVA gain will be adjusted by 0.5dB. You will see the changes in Tx Power and VVA gain as shown in **Figure 3.12**.

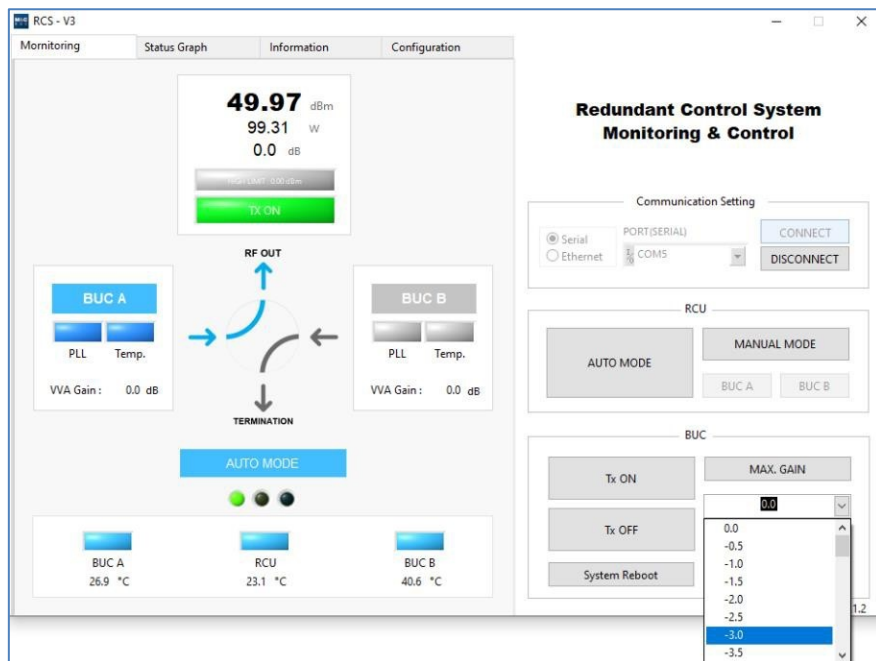


Figure 3.11 – VVA Gain Setting

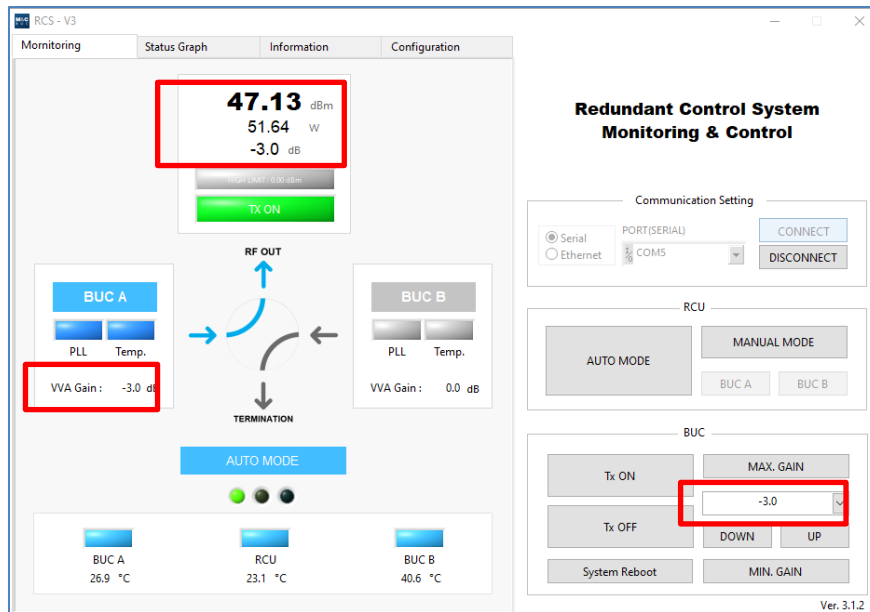


Figure 3.12 – Gain Changed

8) PLL Status [BUC_A / BUC_B]

If the reference source of 10MHz is applied inappropriately, the BUC will be unlocked, and the LED color of PLL Status Indicator will change from blue to red as shown in **Figure 3.13**. If it turns red, check the power level of the 10MHz reference source of the modem side or the BUC input side.



Figure 3.13 – PPL Status [Error]

9) Temperature Status

If the BUC temperature exceeds the shutdown temperature (85°C), the BUC will be shut down automatically, and the LED color of Temperature Status Indicator will change from blue to red as shown in **Figure 3.14**.

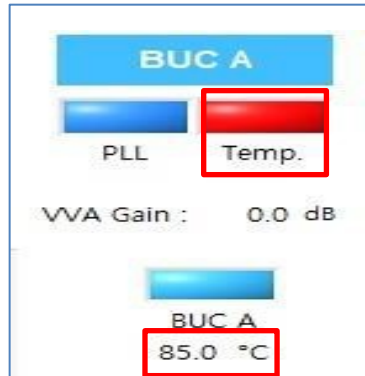


Figure 3.14 - Temperature Status [Error]

When the BUC shut down, turn off the BUC and let it cool down for 5 minutes before operating the BUC again. If the same issue keeps appearing, please contact Norsat for further support.

10) Link Status_1

If there is a communication signal error, the LED color of Link Status Indicator will change from blue to red as shown in **Figure 3.15**.

NOTE: In case of Auto Mode, if the BUC A is disabled, then the BUC B will be activated automatically.

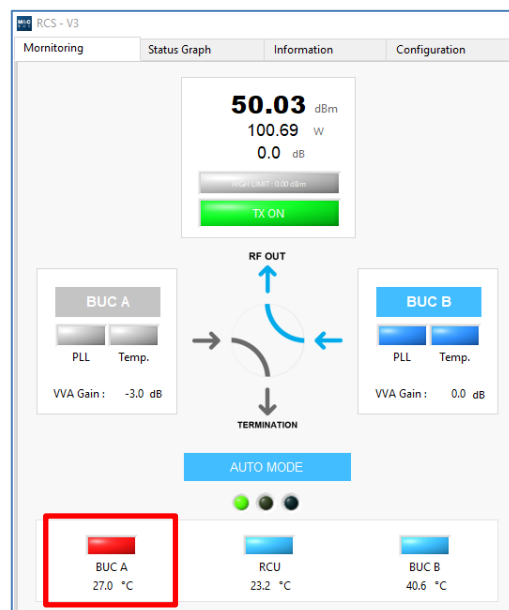


Figure 3.15 - Link Status_1 [Error]

11) Link Status_2

If a communication error occurs on both the BUC A and the BUC B, the LED color of Link Status Indicator will change from blue to red as shown in **Figure 3.16**.

Both BUC A and the BUC B become disabled.

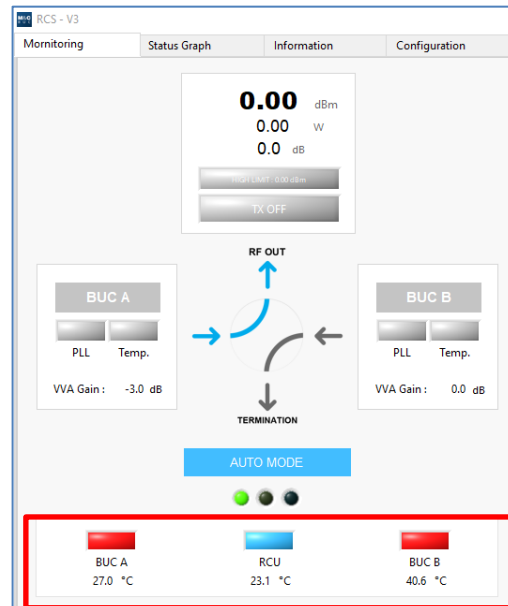


Figure 3.16 - Link Status_2 [Error]

12) Status Graph and Save Data

If you click the Status Graph tab, you can see the BUC status graph as shown in **Figure 3.17**.



Figure 3.17 – Status Graph

- To save the data, click Save box.
- Then, click the folder  icon to set the file location and filename extension. The data (date, time, Tx Power, temperature of the RCU, the BUC A and the BUC B, VVA Gain, Tx On/Off, BUC Select, Auto/Manual, PLL, Temperature, Power High Limit Alarm, and Power High Limit Value) will be saved as shown in **Figure3.18**.
- The graph will be reset if you click the Graph Clear button.

Date	Time	Tx Power	VVA Gain	Temperature -RCU	Temperature -BUC A	Temperature -BUC B	Tx ON/OFF (0:Off / 1:On)	BUC Select (0:BUC A / 1:BUC B)	AUTO/MANUAL (0:AUTO / 1:MANUAL)	PL (0:normal / 1:error)	Temperature (0:normal / 1:error)	Power High Limit Alarm (0:normal / 1:error)	Power High Limit Value
10/24/2022	2:24:54 PM	48.64	0	23.4	26.5	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:05 PM	48.64	0	23.4	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:06 PM	48.65	0	23.4	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:07 PM	48.64	0	23.4	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:08 PM	48.65	0	23.4	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:09 PM	48.66	0	23.4	26.5	39.1	1	0	0	0	0	0	0
10/24/2022	2:25:10 PM	48.65	0	23.5	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:11 PM	48.64	0	23.5	26.4	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:12 PM	48.65	0	23.1	26.6	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:13 PM	48.65	0	23.4	26.4	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:14 PM	48.65	0	23.5	26.4	39.2	1	0	0	0	0	0	0
10/24/2022	2:25:15 PM	48.64	0	23.4	26.4	39.2	1	0	0	0	0	0	0

Figure 3.18 – Saved Data

13) Information

If you click the Information tab, you can see the Log and System Information as show in **Figure 3.19**.

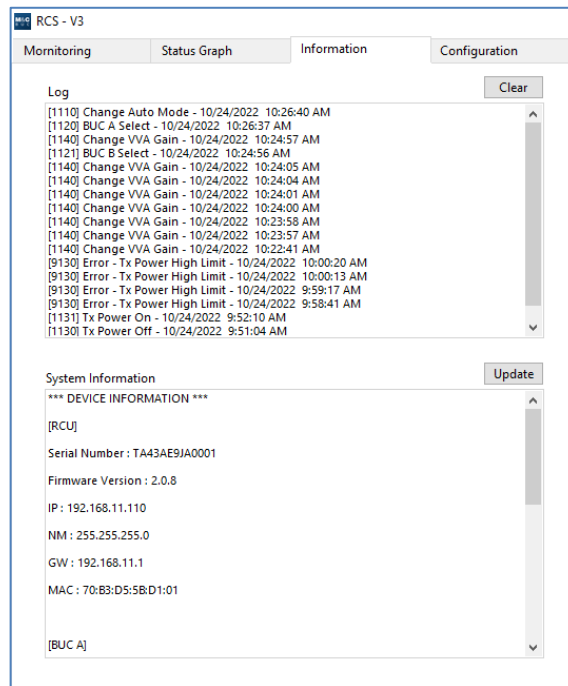


Figure 3.19 - Information

- Logs are shown for the control and error conditions.
Log records are not saved automatically. Click Clear button to clear the record.
- System information displays the information about the RCU, the BUC A, and B.

14) Configuration

If you click the Configuration tab, you can see the default BUC selection, Ethernet port (TCP), BUC address change, and Tx Power High Limit as shown in **Figure 3.20**.

The screenshot shows the 'Configuration' tab of the 'RCS - V3' interface. It contains four configuration panels:

- Main BUC Selection:** Features two radio buttons, 'BUC A' (selected) and 'BUC B'. Below them are 'LOAD' and 'SAVE' buttons.
- Ethernet Port(TCP):** Features a text input field containing the value '0'. Below it are 'LOAD' and 'SAVE' buttons.
- BUC address change:** Features two text input fields labeled 'current' and 'new', separated by a right-pointing arrow. Below them is a 'BUC address change' button. A caution note states: '[CAUTION] Only one BUC should be connected with a M&C cable for this action.'
- Tx Power High Limit:** Features a text input field containing the value '0'. Below it are 'LOAD' and 'SAVE' buttons.

Figure 3.20 – Information

- a. The default setting is the BUC A, but you can change it from the Main BUC Selection.

This close-up shows the 'Main BUC Selection' panel. The 'BUC A' radio button is selected and highlighted with a red rectangular box. The 'BUC B' radio button is unselected. Below the selection area are 'LOAD' and 'SAVE' buttons.

Figure 3.21

- i. Click LOAD to display the current selected BUC.
 - ii. Click SAVE after changing the BUC.
- b. Ethernet Port (TCP) can be set up and modified.

This close-up shows the 'Ethernet Port(TCP)' panel. The text input field contains the value '4200' and is highlighted with a red rectangular box. Below the input field are 'LOAD' and 'SAVE' buttons.

Figure 3.22

The default value for Ethernet port is 4200.

- i. Click LOAD to show the current set value.
- ii. Click SAVE after changing the value.

c. Change BUC address

current

new

Change BUC address

[CAUTION]
Only one BUC should be connected with a M&C cable for this action.

Figure 3.23

After replacing the new BUC, the address should be changed.

 **Caution:** Only one BUC should be connected with a M&C cable for this action.

- Current: Address of BUC to be changed (A-Z, 0-9)
- New: Address of new BUC (A or B)

Use addresses based on the position (BUC A position= A or BUC B position= B).

d. Tx Power High Limit

Tx Power High Limit

0

LOAD SAVE

Figure 3.24

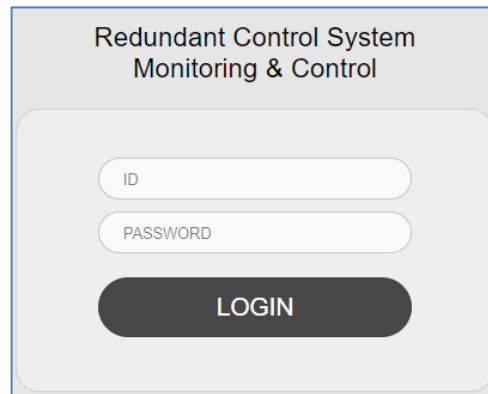
When the Tx power level is higher than the setting value, the High Limit indicator changes to red in the Monitoring tab.



4. Web-Based Monitoring and Control

4.1 Web-Based RCS Operation

-  Do not connect more than five devices or open this webpage at the same time. The webpage can be slow down or stopped.
1. Open any web browser and enter the IP address 192.168.11.XXX in the address bar. The default IP address is 192.168.11.110.
 2. Once the login page shows up as **Figure 4.1**, type in your ID and password. The default ID and PW are admin and admin.



The login page is titled "Redundant Control System Monitoring & Control". It features a light gray background with a white rounded rectangle in the center. Inside this rectangle, there are two input fields: "ID" and "PASSWORD". Below these fields is a dark gray button with the text "LOGIN" in white capital letters.

Figure 4.1 – Login Page

3. Click LOGIN, and the monitor and control page will be connected as shown in **Figure 4.2 & Figure 4.3**.



Figure 4.2 – Monitoring & Control Page 1

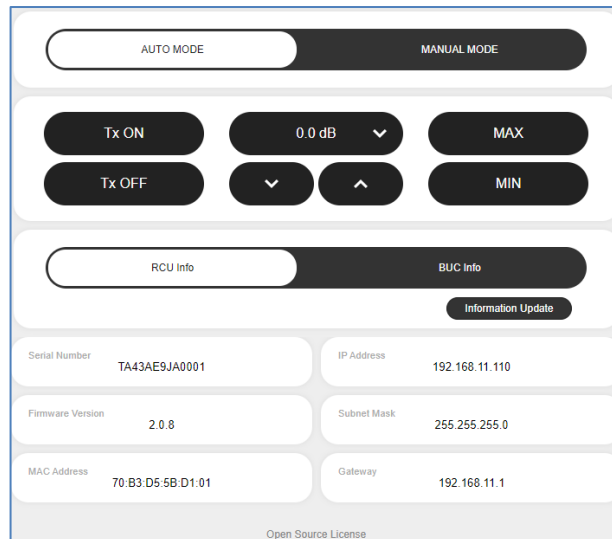


Figure 4.3 – Monitoring & Control Page 2

- ⚠ Please change the password for security purpose once you login.
- ⚠ If you forget the password, please contact Norsat for further support.

4.2 Description of Web-based Monitoring and Control

4.2.1 BUC Status



Figure 4.4

1. Tx Status Indicators

This section shows the Tx Power level, VVA Gain value, Tx power high limit alert status, and Tx ON/OFF status.

2. BUC Status [BUC A & BUC B]

This section shows the on-time status of VVA gain, PLL, temperature of the BUC.

3. Waveguide Switch Status

This section shows the on-time status of waveguide switch path of the BUC to RF OUT.

4.2.2 LED Indicators & Temperature



Figure 4.5

1. shows the RCU to the BUC_A communication status and temperature.
2. shows the PC to the RCU communication status and temperature.
3. shows the RCU to the BUC_B communication status and temperature.

4.2.3 Mode Selection Window



Figure 4.6

1. Auto Mode
If there is a problem with a current operating unit, this mode will switch over to another unit automatically.
2. Manual Mode
If there is a problem with a current operating unit, the operator has to switch over to another unit manually.



Figure 4.7

4.2.4 Control Window

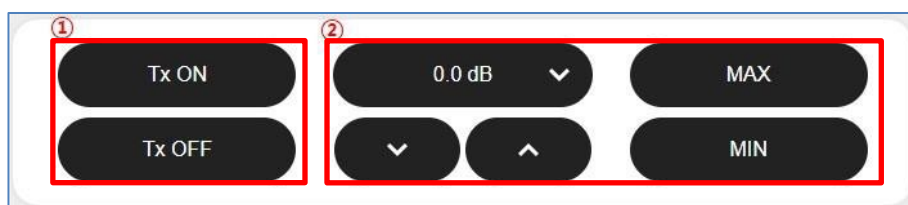


Figure 4.8

1. Tx ON/OFF
These buttons are used to turn on or off the Tx of the BUC.
2. VVA Gain Setting
These buttons are used to change the gain of the BUC.

4.2.5 RCU Information Display

This section shows the information of the RCU including serial number, firmware version, IP address, subnet mask, gateway, and MAC address.

RCU Info

BUC Info

Information Update

Serial Number	TA43AE9JA0001	IP Address	192.168.11.110
Firmware Version	2.1	Subnet Mask	255.255.255.0
MAC Address	70:B3:D5:5B:D1:01	Gateway	192.168.11.1

Figure 4.9

When you Click the information Update button, the updated information will be applied to this page.

4.2.6 BUC Information Display

This section shows the information of the BUC including serial number, firmware version, power class, frequency band, and frequency.

RCU Info

BUC Info

Information Update

BUC A		BUC B	
Serial Number	TA46CA9KT0001	Serial Number	TA46CA9JT0002
Firmware Version	4.0	Firmware Version	4.0
Power Class	40	Power Class	40
Frequency Band	C Band	Frequency Band	C Band
Frequency	6725MHz ~ 7025MHz	Frequency	6725MHz ~ 7025MHz

Figure 4.10

4.3 Operation

- ⓘ Connect the RCU and the PC with an Ethernet cable and ensure that they are connected appropriately.
- ⓘ This webpage does show the real-time operating status.

 If the RCU has any defects, turn off the power of the deactivated BUC before replacing the new RCU.

4.3.1 Monitoring

1. The BUC will be turned on by default with green light as shown in **Figure 4.11**.



Figure 4.11 – Webpage Default Status Without IF Signal

2. Tx ON/OFF Status

If you click Tx ON or OFF button, the ON/OFF status indicator of the BUC will turn green or gray as shown in **Figure 4.12**.

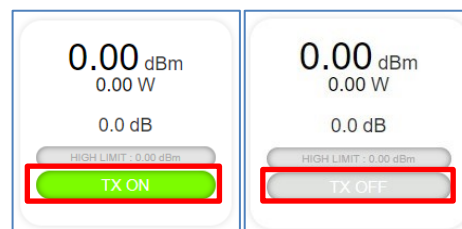


Figure 4.12 – 23. Tx ON and OFF

3. VVA Gain Setting

To adjust the gain, select the values from the drop down menu from -20 dB to 0 dB as shown in **Figure 4.13**. Or it can be adjusted by clicking ^ (UP), v (DOWN), MAX, or MIN buttons. If you click ^ (UP) or v (DOWN) button, the VVA gain will be adjusted by 0.5dB. You will see the changes in the Tx Power and VVA gain as shown in **Figure 4.14**.

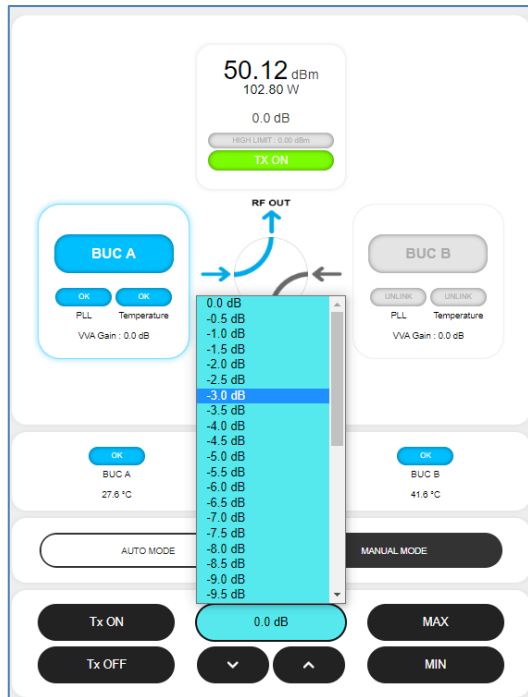


Figure 4.13 – VVA Gain Setting



Figure 4.14 – Gain Changed

4. PLL Status

If the reference source of 10MHz is applied inappropriately, the BUC will be unlocked, and the LED color of PLL status indicator will change from blue to red as shown in **Figure 4.15**. If it turns red, check the power level of the 10MHz reference source of the modem side or the BUC input side.



Figure 4.15 – PLL Status [Error]

5. Temperature Status

If the BUC temperature exceeds the shutdown temperature, the BUC will shut down automatically, and the LED color of Temperature status indicator will change from blue to red as shown in **Figure 4.16**.



Figure 4.16 – Temperature Status [Error]

When the BUC shut down, turn off the BUC and let it cool down for 5 minutes before operating the BUC again. If the same issue keeps appearing, please contact to Norsat for further support.

NOTE: In case of Auto Mode, if the BUC A is disabled, then the BUC B will be activated automatically. In case of manual mode, the operator must switch over to another unit manually.

6. Link Status

If there is a communication signal error, the LED color of Link Status indicators will change from blue to red as shown in **Figure 4.17**.

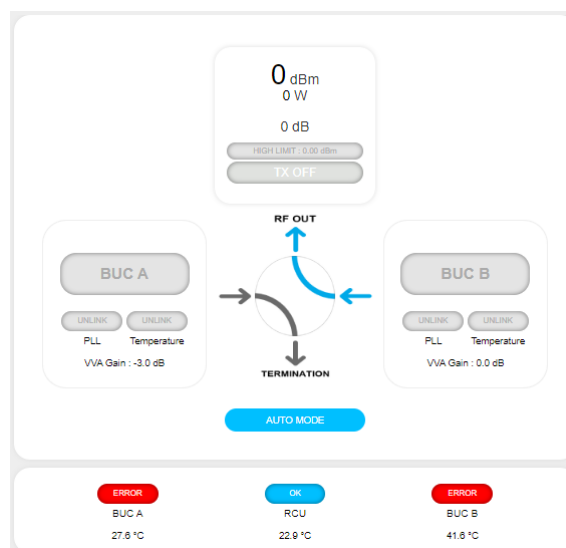


Figure 4.17 - Link Status [Error]

- 1) shows the RCU to the BUC A communication status.
- 2) shows the PC to the RCU communication status.
- 3) shows the RCU to the BUC B communication status.

7. Status Graph

If you click the STATUS GRAPH tab, you can see the RCU & the BUC status graph as shown in **Figure 4.18**.

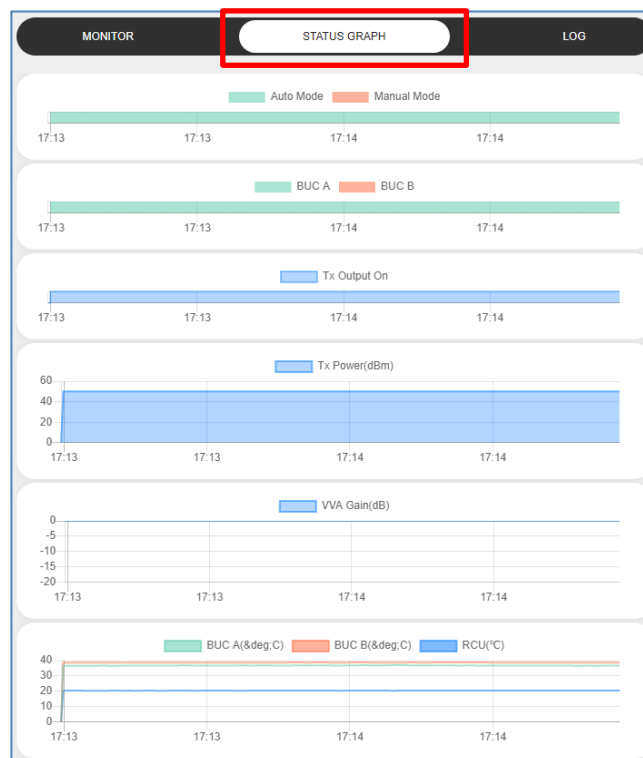


Figure 4.18 – Status Graph

8. LOG

If you click the LOG tab, you can see the RCU and the BUC control history as shown in **Figure 4.19**.

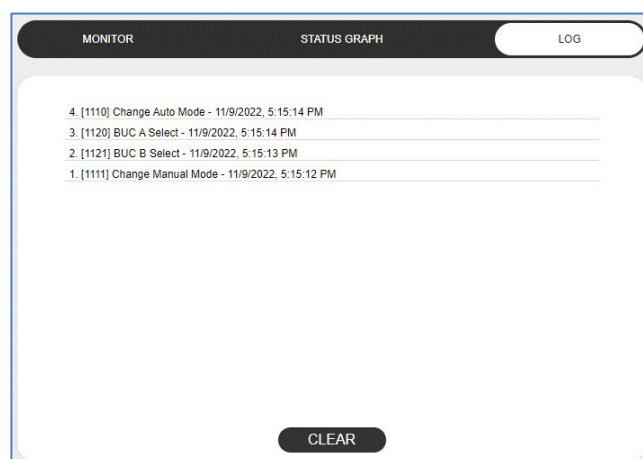


Figure 4.19 - LOG

4.3.2 Menu Bar

If you click the Menu button, you can see the menu bar (Admin / Network /Config /SNMP / Upgrade / Logout) as shown in **Figure 4.20**.



Figure 4.20 – Menu Button and Menu Bar

4.3.2.1 Admin

1. To change the password, click Admin from the menu bar.
2. The password change page will appear as shown in **Figure 4.21**.
3. Enter ID, old password and new password, then click CHANGE button.
4. The password can be up to 16 characters.

* Default ID: admin, Default Password: admin

Figure 4.21 – Password Change Page

4.3.2.2 Network

1. To change IP address, subnet mask, and gateway, click Config from the menu bar.
2. Change Network Configuration page appears as shown in **Figure 4.22**.

Figure 4.22 - Change Network Configuration Page

3. Type the numbers into each blank, then click CHANGE button.

- Default IP Address / Subnet mask / Gateway are:
- IP Address: 192.168.11.110
- Subnet mask: 255.255.255.0
- Gateway: 192.168.11.1

4. Enter the changed IP address in the address bar of the browser.

4.3.2.3 Config

To change the selected BUC and TCP (Port), click Config from the menu bar. Change Configuration page appears as shown in **Figure 4.23**.

Figure 4.23 - Change Configuration Page

- Main BUC Select**
Select the BUC used as the default when the system restarts
- TCP Port for control program**
TCP port number for connection to be used in control program (NGW RCS)
- Tx power high limit**
When the Tx power level is higher than the setting value, the high limit indicator changes to red in the Monitoring tab.
- BUC address change**
Change the address of the BUC after replacing the BUC.

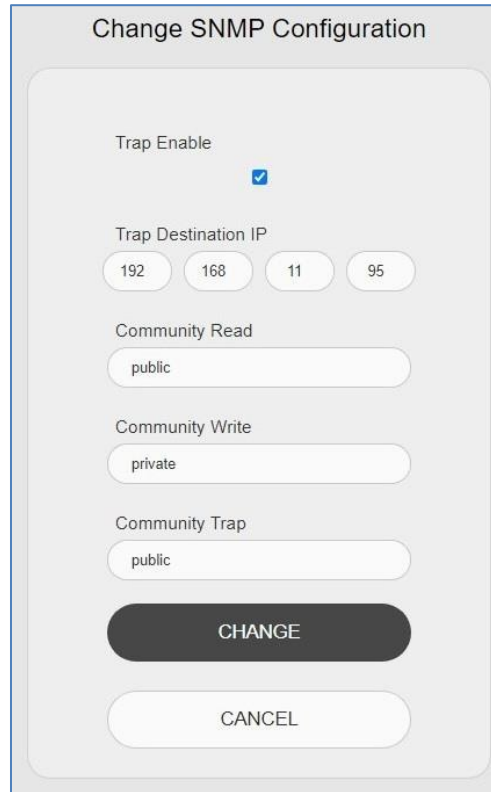
⚠ Caution: Only one BUC should be connected with a M&C cable for this action.

- Current: Address of BUC to be changed (A-Z, 0-9)
- New: Address of new BUC (A or B)

Use addresses based on the position (BUC A position= A or BUC B position= B)

4.3.2.4 SNMP

1. To change the SNMP configuration, click SNMP from the Menu bar.
2. Change SNMP Configuration page appears as shown in **Figure 4.24**.

The image shows a web interface titled "Change SNMP Configuration". It contains several configuration fields: "Trap Enable" with a checked checkbox, "Trap Destination IP" with four input fields containing "192", "168", "11", and "95", "Community Read" with a text field containing "public", "Community Write" with a text field containing "private", and "Community Trap" with a text field containing "public". At the bottom, there are two buttons: "CHANGE" and "CANCEL".

Change SNMP Configuration

Trap Enable ☒

Trap Destination IP

192 168 11 95

Community Read

public

Community Write

private

Community Trap

public

CHANGE

CANCEL

Figure 4.24 – Change SNMP Configuration Page

3. To enable or disable the SNMP traps, check or uncheck the box.
4. Enter the IP address where SNMP traps will be sent.
5. Enter the SNMP v2C password into the community Read-Write and Trap. The password can be up to 16 characters.
6. Click CHANGE button once all information is entered.

⚠ Please change the password for security purpose once you login.

⚠ If you forget the password, please contact Norsat for further support.

4.3.2.5 Upgrade

1. RCU Firmware Upgrade

⚠ DO NOT SHUT DOWN the RCU and the PC or refresh the webpage in the middle of the upgrade installation. It may cause serious damage to the firmware. If the RCU or the PC shuts down due to the power failure, then wait for some time and restart the upgrade. If the webpage or the RCU does not work properly, please contact Norsat for further support. During the upgrade, the RCU will not function properly because the mode changes and the upgrade progresses.

⚠ To upgrade the firmware, upgrade with **ONLY one PC** connecting with the RCU.

- 1) To upgrade the firmware, download the upgrade file from the Norsat website or request the file to support@norsat.com.
- 2) Click Upgrade button from the menu bar.
- 3) Click OK in the popup window to proceed with the upgrade as shown in **Figure 4.25**.

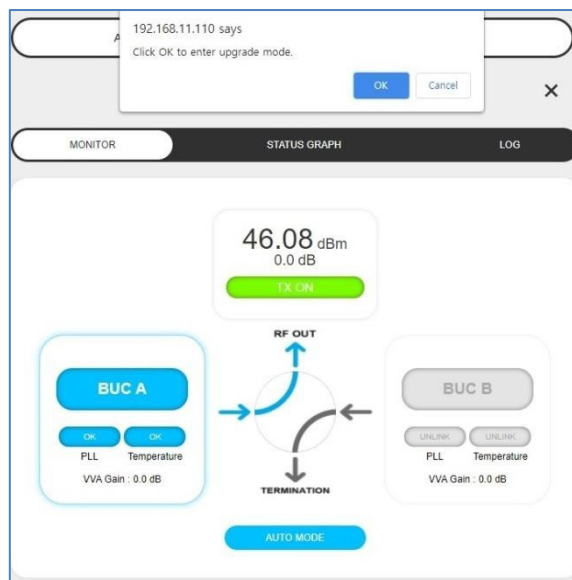


Figure 4.25 – Upgrade Popup Window

- 4) Wait for a minute once Enter Upgrade Mode page appears as shown in **Figure 4.26**.

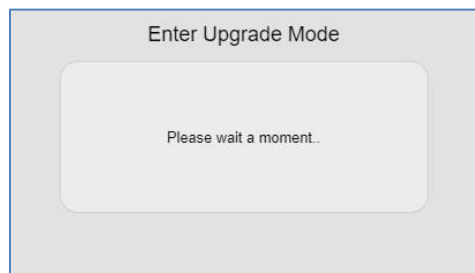


Figure 4.26 - Webpage Enter Upgrade Mode Page

- 5) Click Choose File button and select the upgrade file (NGW_RCU_X.X.X.bin) as shown in **Figure 4.27**.

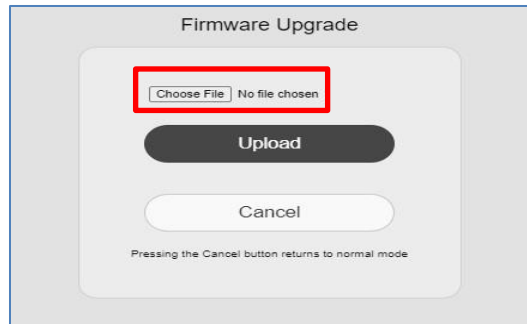


Figure 4.27 – Firmware Upgrade Page

- 6) Once you click Upload, the upgrade process starts immediately.

During the upgrade, the BUC will not work properly. If you do not want to stop the BUC, click Cancel button before upgrading the firmware as shown in **Figure 4.28**.

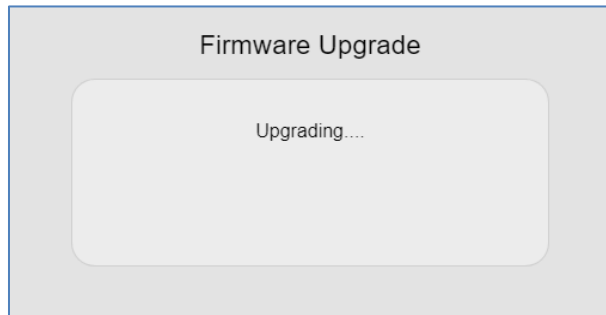


Figure 4.28 – Firmware Upgrading

- 7) Once the upgrade has been completed, the Enter Normal Mode page will be shown up and directly connects to the login page as shown in **Figure 4.29** and **Figure 4.30**.

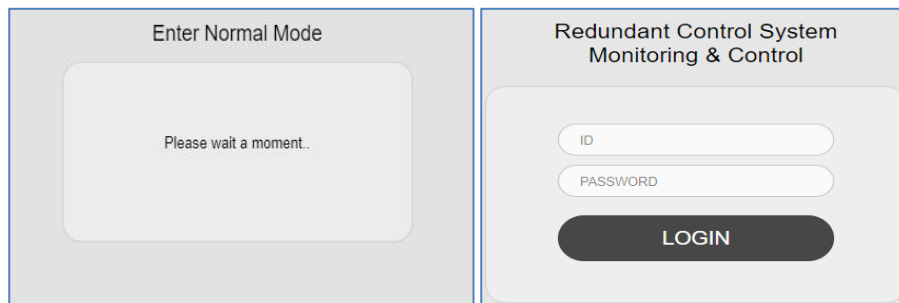


Figure 4.29 – Enter Normal Mode Page Figure

Figure 4.30 – Login Page

- 8) If the Monitor and Control webpage does not load properly after login, refresh the page or close and reopen the browser.
2. BUC A / BUC B Firmware Upgrade
- 1) Required Items
- BUC upgrade cable
 - RCU firmware file (NGW_BUC_x.x.x.bin)

To upgrade the firmware, download the upgrade file from the Norsat website or request the file to support@norsat.com.

- 2) Follow the **Figure 4.31** to connect the PC and the BUC using a [PC to RCU cable (Ethernet) + BUC Upgrade cable].

Refer to the pin composition of [RCU] port on **Figure 4.32** below.

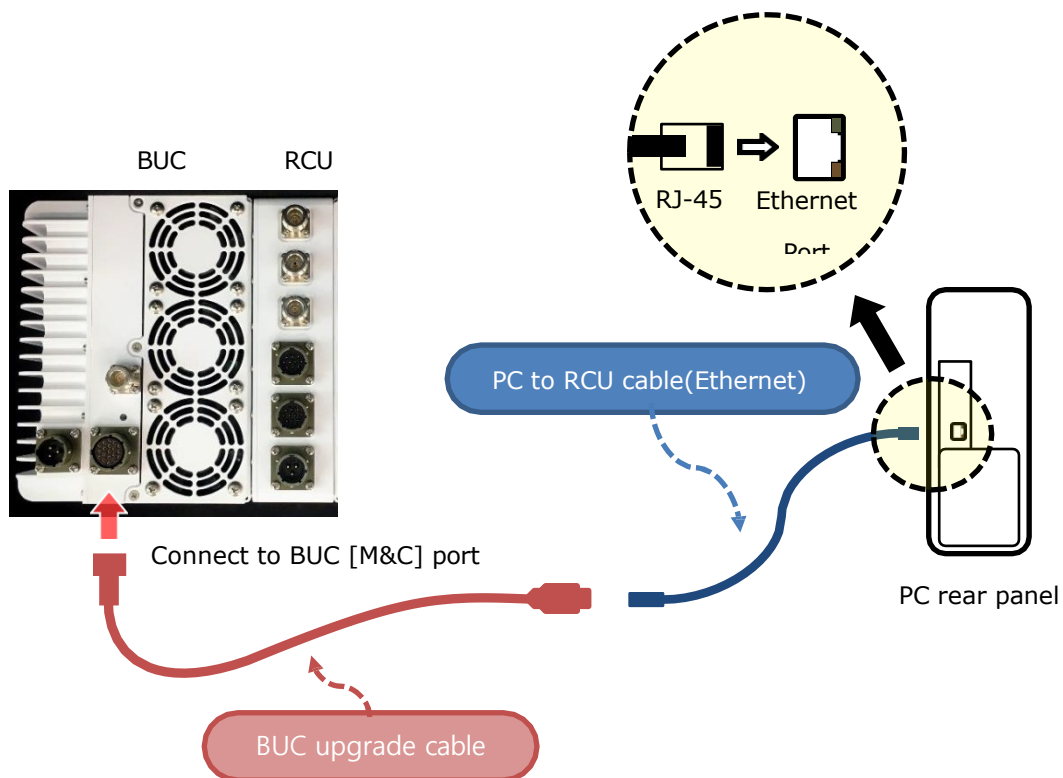


Figure 4.31 – Connection Between PC and BUC

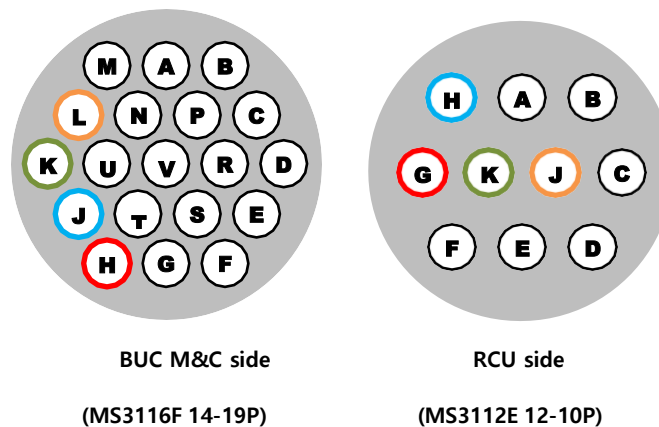


Figure 4.32 – Physical Interface of Connector of the BUC Upgrade Cable

BUC M&C side (MS3116F 14-19P)	Name	RCU side (MS3112E 12-10P)
H	Ethernet Rx+	G
J	Ethernet Rx-	H
K	Ethernet Tx+	K
L	Ethernet Tx-	J

Table 4.1

3) Upgrade

- ⚠ DO NOT SHUT DOWN the BUC and the PC or refresh the webpage in the middle of the upgrade installation. It may cause serious damage to the firmware. If the BUC or the PC shuts down due to the power failure, then wait for some time and restart the upgrade. If the webpage or the BUC does not work properly, please contact Norsat for further support. During the upgrade, the BUC will not function properly because the mode changes and the upgrade progresses.
- ⚠ To upgrade the firmware, upgrade with **ONLY one PC connecting with the RCU.**
- a. Open any web browser and enter the IP address 192.168.11.XXX in the address bar. The default IP address is 192.168.11.101.
- b. Once the login page shows up as **Figure 4.33**, type in your ID and password. The default ID and PW are admin and admin.

The image shows a web browser window displaying a login page. The page has a light gray background with a darker gray header area containing the text 'Monitoring and Control'. Below the header is a white rounded rectangle containing the login form. The form has two input fields: 'ID' and 'PASSWORD', both with placeholder text. Below these fields is a dark gray button with the text 'LOGIN' in white.

Figure 4.33 – Login Page

- c. Click LOGIN, and the MONITORING & CONTROL page will show up as shown in **Figure 4.34**.

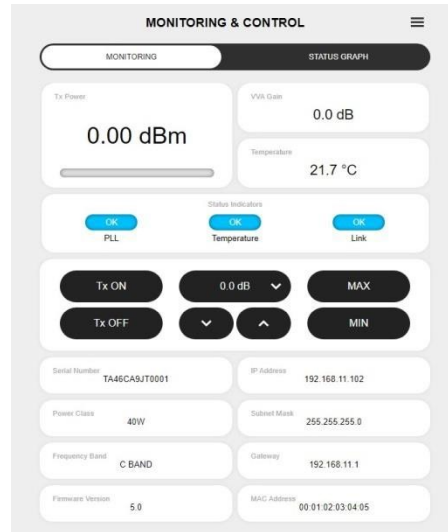


Figure 4.34 – Monitoring & Control Page

- d. Click UPGRADE from the menu bar

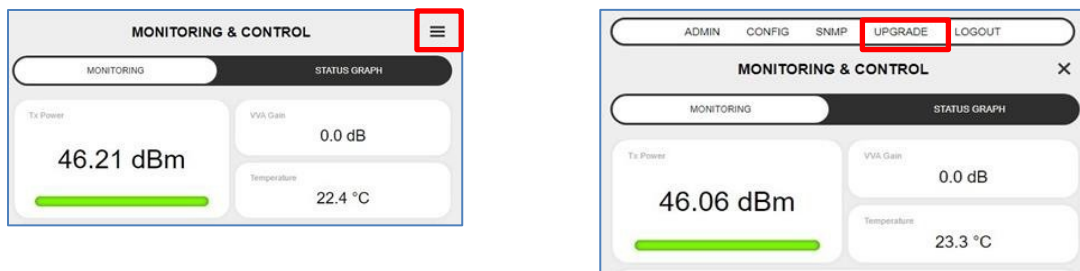


Figure 4.35 – Menu Button and Menu Bar

- e. Click OK in the popup window to proceed with the upgrade.

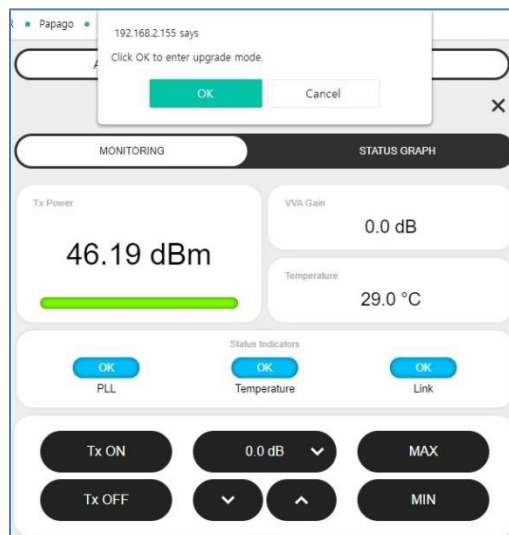


Figure 4.36 – Upgrade Popup Window

- f. Wait for a minute once Enter Upgrade Mode page appears.

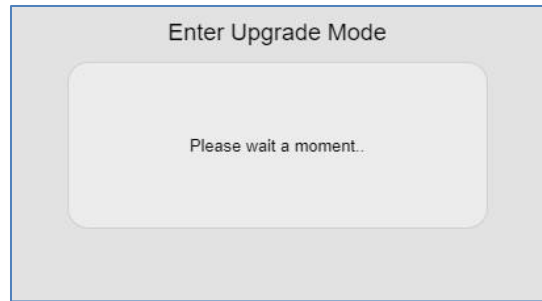


Figure 4.37 - Webpage Enter Upgrade Mode Page

- g. Click Choose File and select the upgrade file (NGW_BUC_x.x.x.bin).

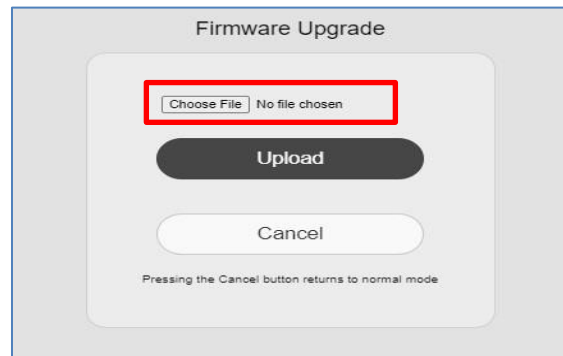


Figure 4.38 – Firmware Upgrade Page

- h. Once you click Upload, the upgrade process starts immediately.
During the upgrade, the BUC will not work properly. If you do not want to stop the BUC, click Cancel before upgrading the firmware.

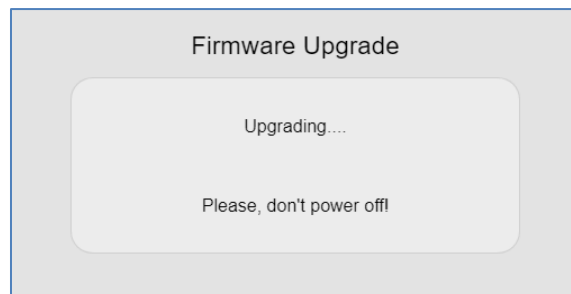


Figure 4.39 – Firmware Upgrading

- i. Once the upgrade is complete, Enter Normal Mode page will show up and directly connect to the login page as shown in **Figure 4.40** and **Figure 4.41**.

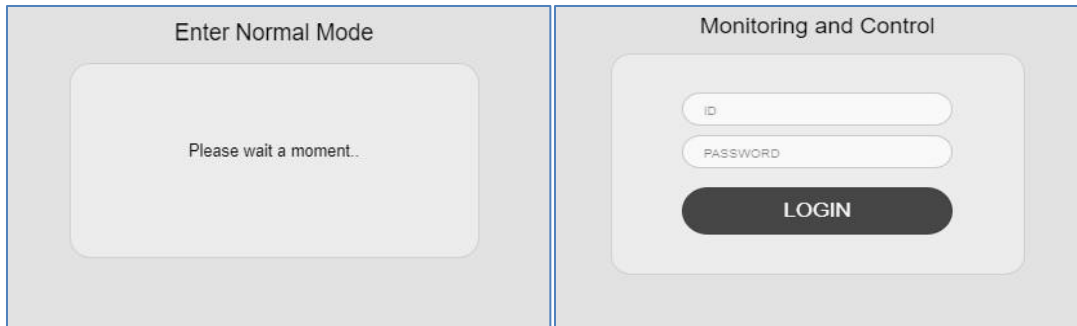


Figure 4.40 – Enter Normal Mode Page

Figure 4.41 Login Page

If the Monitor and Control webpage does not load properly after login, refresh the page or close and reopen the browser.

- j. To upgrade the firmware of another BUC, detach the connector connected with the BUC [M&C] port and connect it with another [M&C] port of BUC, then repeat the steps of ①-⑨ above to upgrade the firmware.
- k. Once the upgrade is completed, detach the [BUC upgrade cable] and set aside and connect the [PC to RCU Cable (Ethernet)] with [RCU] port of the RCU.

4.3.2.6 Logout

If you want to logout from the webpage, click Logout from the menu bar.

5. Customer Service

5.1 Warranty Repair

Norsat International Inc. will repair defective parts, free of charge, for TWO (2) YEARS from the date of its shipment from the Norsat factory.

During the warranty year, the cost of shipment to and from Norsat factory will be covered. Please note that if the equipment is not found to be defective, the client will be responsible for the cost of testing and shipping the item.

5.2 Exceptions to Warranty

Norsat International Inc. shall have no liability or obligation to purchaser with respect to the product subjected to any of the following:

1. Opened unit
2. The unit exposed to possible damaging situations or those that are not manipulated under the security conditions described in the user manual
3. Failure caused by operation of the product outside its published electrical or environmental specifications, or any causes other than ordinary use

5.3 Other Repairs

Non-warranty and out-of-warranty repair service is available from Norsat International Inc. The repair service can be obtained by contacting Norsat International Inc. And a Return Material Authorization (RMA) number must be obtained before returning.

5.4 Contact Information

Address: 4020 Viking Way Unit 110
Richmond, British Columbia
V6V 2N2 Canada

Phone: +1 604-821-2800



ABOUT NORSAT

Norsat International Inc., founded in 1977, is a leading provider of innovative communication solutions that enable the transmission of data, audio and video for remote and challenging applications. Norsat's products and services include customizable satellite components, portable satellite terminals, maritime solutions and satellite networks. The company's products and services are used extensively by telecommunications services providers, emergency services and homeland security agencies, military organizations, health care providers and Fortune 1000 companies.

110 – 4020 Viking Way | Richmond | British Columbia | Canada

V6V 2N2 | support@norsat.com

www.norsat.com